Deep Generative Models

Chapter 4: Generative Adversarial Networks

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Questioning Validity of GAN

+ *Though intuitive, is there any ways to show that this training approach is indeed leading to sampling from data distribution?*
– Well! We can claim this, if we can show the following

> *Let $G_{\mathbf{w}^\star}$ and $D_{\phi^\star}$ be solutions to the min-max game; then, $x = G_{\mathbf{w}^\star}(z)$ for $z \sim Q(z)$ is approximately distributed by data distribution*

To show this arguments let's consider the following idealistic assumptions

- *Discriminator D_ϕ can realize any function D exactly*
- *Generator $G_{\mathbf{w}}$ can realize any mapping G exactly*
  - ↳ *Discriminator and generator are infinitely-deep NNs*
- *Dataset $\mathbb{D}$ has infinitely large number of samples*
  - ↳ *Our estimated average is accurate, i.e., $\hat{\mathbb{E}} \rightsquigarrow \mathbb{E}$*

Min-Max Game in *Idealistic Setting*

With an ideal setting: *the objective of the min-max problem becomes*

$$\mathcal{L}(G, D) = \mathbb{E}_{x \sim P}\{\log D(x)\} + \mathbb{E}_{z \sim Q}\{\log(1 - D(G(z)))\}$$

Since we assume G is any generator: we can say

$x = G(z)$ *can be distributed by any* $\hat{P}$ *on* $\mathbb{X}$

↳ *By changing* $G \rightsquigarrow \hat{P}$ *is changed*[a]

[a] *Recall that* $\hat{P}$ *is not necessarily computable!*

With these definitions, we could say

$$\begin{aligned}\mathcal{L}(G, D) &= \mathbb{E}_{x \sim P}\{\log D(x)\} + \mathbb{E}_{x \sim \hat{P}}\{\log(1 - D(x))\} \\ &\equiv \mathcal{L}\left(\hat{P}, D\right)\end{aligned}$$

Optimal Players: *Discriminator*

Let's start with finding *theoretically-optimal generator and discriminator: we could expand the objective as*

$$\begin{aligned}\mathcal{L}\left(\hat{P}, D\right) &= \mathbb{E}_{x\sim P}\left\{\log D\left(x\right)\right\} + \mathbb{E}_{x\sim \hat{P}}\left\{\log\left(1 - D\left(x\right)\right)\right\} \\ &= \int_{\mathbb{X}} \log D\left(x\right) P\left(x\right)\mathrm{d}x + \int_{\mathbb{X}} \log\left(1 - D\left(x\right)\right) \hat{P}\left(x\right)\mathrm{d}x \\ &= \int_{\mathbb{X}} \log D\left(x\right) P\left(x\right) + \log\left(1 - D\left(x\right)\right) \hat{P}\left(x\right)\mathrm{d}x\end{aligned}$$

The min-max game first maximizes $\mathcal{L}$ over D: we can find $D^\star$ by setting

$$\frac{\mathrm{d}}{\mathrm{d}D}\mathcal{L}\left(\hat{P}, D\right) \stackrel{!}{=} 0$$

+ *How can we compute derivative with respect to a function?!*
– *We should use Radon-Nikodym derivative, but let us do it naively!*

Optimal Discriminator: *Naive Analysis*

! Attention

Following analysis is simplified and not accurate, but reflects key idea

Under some regularity conditions, we can find the optimal D by writing

$$\frac{\mathrm{d}}{\mathrm{d}D}\mathcal{L}\left(\hat{P}, D\right) = \frac{\mathrm{d}}{\mathrm{d}D}\int_{\mathbb{X}} \log\left(D\left(x\right)\right) P\left(x\right) + \log\left(1 - D\left(x\right)\right) \hat{P}\left(x\right)\mathrm{d}x$$

$$= \int_{\mathbb{X}} \frac{\mathrm{d}}{\mathrm{d}D\left(x\right)} \left[\log\left(D\left(x\right)\right) P\left(x\right) + \log\left(1 - D\left(x\right)\right) \hat{P}\left(x\right)\right] \mathrm{d}x$$

and setting the integrand to zero at every x

$$\frac{\mathrm{d}}{\mathrm{d}D\left(x\right)} \left[\log\left(D\left(x\right)\right) P\left(x\right) + \log\left(1 - D\left(x\right)\right) \hat{P}\left(x\right)\right] = 0$$

$$\frac{P\left(x\right)}{D\left(x\right)} - \frac{\hat{P}\left(x\right)}{1 - D\left(x\right)} = 0$$

Optimal Discriminator: *Naive Analysis*

This conclude that

$$D^{\star}(x) = \frac{P(x)}{P(x) + \hat{P}(x)}$$

is the optimal discriminator that maximizes the objective

Attention

Note that $D^{\star}(x)$ *is not computable!*

- ↳ *We have no access to the data distribution*
- ↳ *We cannot compute explicitly* $\hat{P}(x)$

But, if the discriminator model is deep enough and we have a large number of samples $\rightsquigarrow$ *we can approximate it by a computational model* D_{ϕ}

Optimal Generator

To find optimal generator: *we can use* $D^\star$ *and solve outer minimization, i.e.,*

$$\min_{\hat{P}} \max_{D} \mathcal{L}\left(\hat{P}, D\right) = \min_{\hat{P}} \mathcal{L}\left(\hat{P}, D^\star\right)$$

This means that the optimal generator distribution $\hat{P}$ *is a solution to*

$$\min_{\hat{P}} \mathcal{L}\left(\hat{P}, D^\star\right) = \min_{\hat{P}} \mathbb{E}_{x\sim P}\left\{\log D^\star\left(x\right)\right\} + \mathbb{E}_{x\sim \hat{P}}\left\{\log\left(1 - D^\star\left(x\right)\right)\right\}$$

This reduces to

$$\min_{\hat{P}} \mathbb{E}_{x\sim P}\left\{\log \frac{P\left(x\right)}{P\left(x\right) + \hat{P}\left(x\right)}\right\} + \mathbb{E}_{x\sim \hat{P}}\left\{\log\left(1 - \frac{P\left(x\right)}{P\left(x\right) + \hat{P}\left(x\right)}\right)\right\}$$

Optimal Generator: *Divergence Minimizer*

Average Distribution

Let us define the average of model and data distribution as

$$A_{P,\hat{P}}(x) = 0.5\left(P(x) + \hat{P}(x)\right)$$

which is still a distribution defined on $\mathbb{X}$

With this definition: *we could write the optimization as*

$$\min_{\hat{P}} \mathbb{E}_{x\sim P}\left\{\log \frac{P(x)}{2A_{P,\hat{P}}(x)}\right\} + \mathbb{E}_{x\sim \hat{P}}\left\{\log \frac{\hat{P}(x)}{2A_{P,\hat{P}(x)}}\right\}$$

$$= \min_{\hat{P}} \mathbb{E}_{x\sim P}\left\{\log \frac{P(x)}{A_{P,\hat{P}}(x)}\right\} + \mathbb{E}_{x\sim \hat{P}}\left\{\log \frac{\hat{P}(x)}{A_{P,\hat{P}}(x)}\right\} - 2\log 2$$

Optimal Generator: *Divergence Minimizer*

Recall: *KL Divergence*

The KL divergence between P and Q is defined as

$$D_{\mathrm{KL}}\left(P\|Q\right)=\mathbb{E}_{x\sim P}\left\{\log\frac{P\left(x\right)}{Q\left(x\right)}\right\}$$

We can thus say that

$$\begin{aligned}&\min_{\hat{P}}\mathbb{E}_{x\sim P}\left\{\log\frac{P\left(x\right)}{A_{P,\hat{P}}\left(x\right)}\right\}+\mathbb{E}_{x\sim\hat{P}}\left\{\log\frac{\hat{P}\left(x\right)}{A_{P,\hat{P}}\left(x\right)}\right\}\\&=\min_{\hat{P}}\boxed{D_{\mathrm{KL}}\left(P\|A_{P,\hat{P}}\right)+D_{\mathrm{KL}}\left(\hat{P}\|A_{P,\hat{P}}\right)}\end{aligned}$$

divergence metric

Jensen-Shannon Divergence

Jensen-Shannon Divergence

The Jensen-Shannon divergence between P and $\hat{P}$ is defined as

$$D_{\mathrm{JS}}\left(P\|\hat{P}\right) = D_{\mathrm{KL}}\left(P\|A_{P,\hat{P}}\right) + D_{\mathrm{KL}}\left(\hat{P}\|A_{P,\hat{P}}\right)$$

In Assignment 2 *we played around with JS divergence to see the following*

KL Divergence vs JS Divergence

Under some regularity conditions on P and $\hat{P}$, we can say that

$$D_{\mathrm{JS}}\left(P\|\hat{P}\right) = 0 \leftrightsquigarrow D_{\mathrm{KL}}\left(P\|\hat{P}\right) = 0 \leftrightsquigarrow P = \hat{P}$$

Optimal Generator: *Divergence Minimizer*

In GAN training, we find a generator distribution $\hat{P}$ *is*

$$\hat{P}^{\star} = \underset{\hat{P}}{\operatorname{argmin}}\, D_{\mathrm{JS}}\left(P \| \hat{P}\right) \leftrightsquigarrow \hat{P}^{\star} = \underset{\hat{P}}{\operatorname{argmin}}\, D_{\mathrm{KL}}\left(P \| \hat{P}\right)$$

In other words, as we train using min-max game

distribution of generated sample $x = G(z)$ *tends to data distribution*

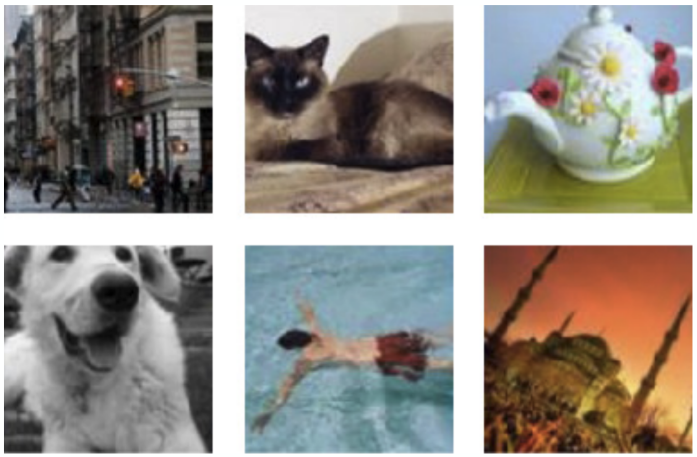
Moral of Story: *Implicit MLE via GAN*

Using min-max training strategy of GAN we implicitly train a generator with minimal KL divergence $\equiv$ *maximal likelihood at its output*

Sample Outputs of GAN

Trained on the dataset

GAN is able to sample



Vanilla GAN: *Stability Challenge*

Vanilla GAN suffers from various practical issues

- *Training loop is unstable*
  - ↳ *Since we use gradient descent for min-max it could easily diverge*
- *Mode collapse happens frequently*
  - ↳ *Generator gives a few good samples all the time to fool discriminator*
  - ↳ *This results in lack of diversity*
- *Computational vanilla GANs are vulnerable to vanishing gradient*
  - ↳ *Objective function returns small gradients*

To overcome these issues, Wasserstein GAN was proposed